Design of a Binary to Gray Code Converter

Digital Design and Computer Architecture (DDCA) | B.Tech 1st Year | Final Project Demonstration

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Section

[A]

Introduction and Objective

What is Gray code?

- A binary code where two consecutive numbers differ in only one bit
- Plain binary can flip many bits at once: 0111 to 1000 flips all 4
- In Gray code the same step is 0100 to 1100, only 1 bit changes
- Fewer changing bits means fewer errors in real hardware

Example

Binary 0101 → Gray 0111

Project Objective

1. **Design**
   a 4-bit Binary to Gray code converter using only logic gates
2. **Build**
   the circuit in Logisim and test every input from 0000 to 1111
3. **Verify**
   the outputs against the truth table and the Boolean equations

Theory and Logic Circuit

Conversion rule

Copy the MSB. Each other Gray bit = XOR of that binary bit and the bit above it.

Boolean equations

$$G3 = B3$$

$$G2 = B3 \oplus B2$$

$$G1 = B2 \oplus B1$$

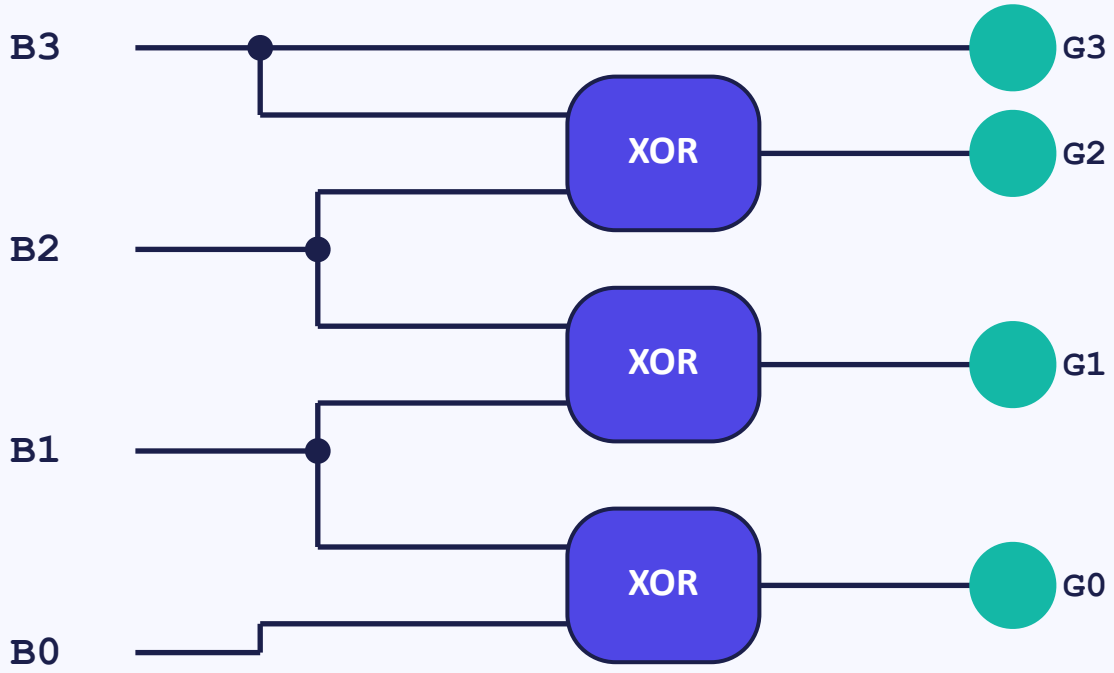
$$G0 = B1 \oplus B0$$

A	B	A⊕B
0	0	0
0	1	1
1	0	1
1	1	0

XOR = 1 if inputs differ

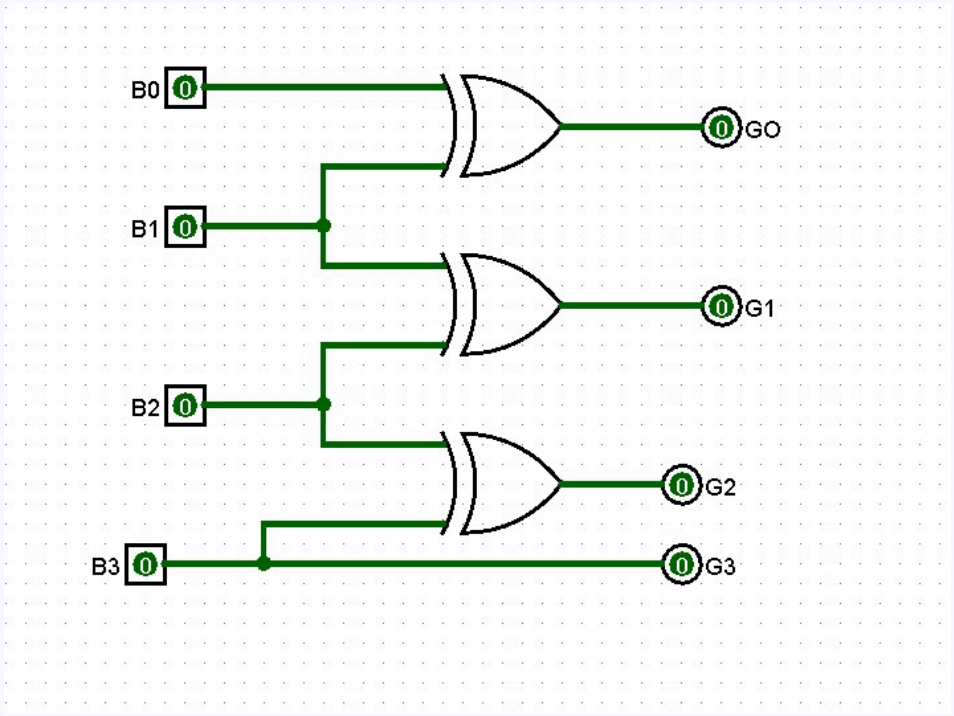
Try 1011: 1, 1⊕0=1, 0⊕1=1, 1⊕1=0 → 1110

Circuit diagram (3 XOR gates)



Inputs on the left, Gray outputs on the right. B3 goes straight to G3.

Circuit Design in Logisim



Components used

4	Input pins
3	XOR gates
4	Output LEDs

How the circuit works

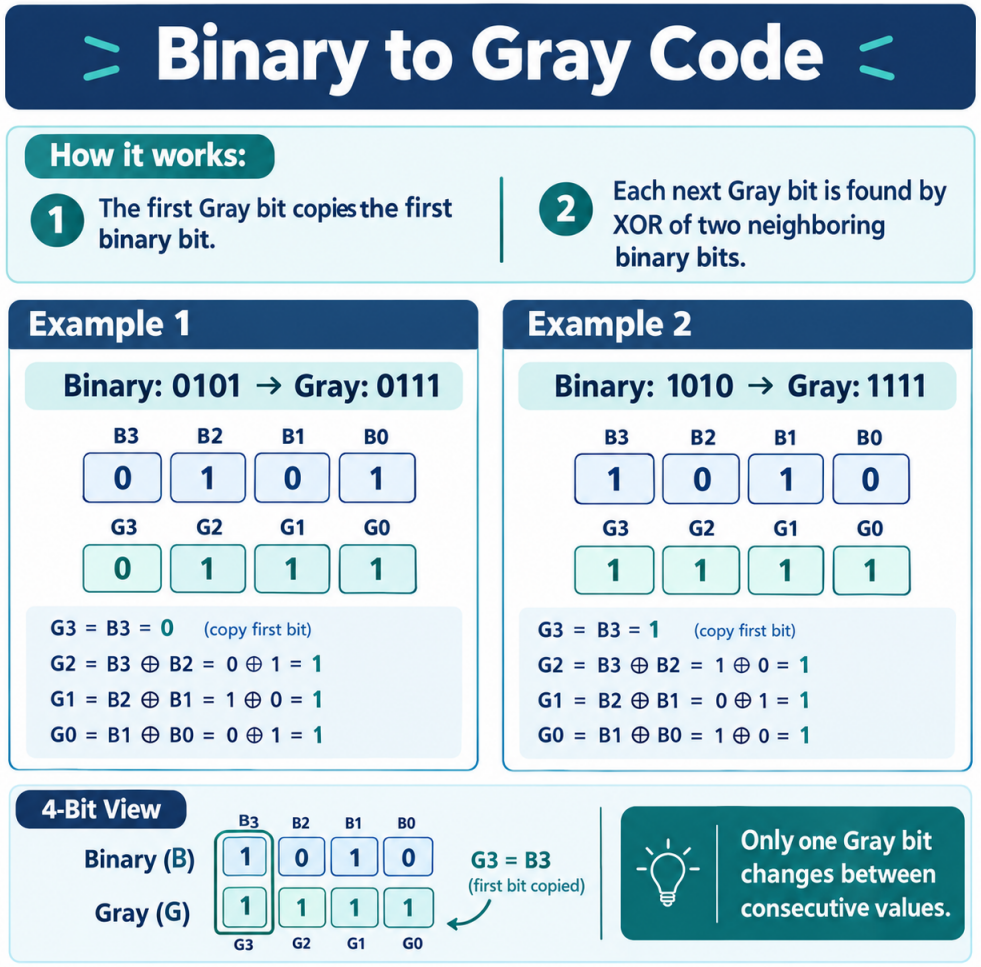
1. **Set the inputs**
   Click the 4 input pins to choose B3 B2 B1 B0
2. **XOR gates work**
   Each gate compares two neighbouring bits
3. **Direct wire**
   B3 is connected straight to G3
4. **Read the output**
   LEDs light up to show G3 G2 G1 G0

Tip: use the Poke tool to toggle the inputs

Simulation Results

Dec	B3	B2	B1	B0	G3	G2	G1	G0
0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	1
2	0	0	1	0	0	0	1	1
3	0	0	1	1	0	0	1	0
4	0	1	0	0	0	1	1	0
5	0	1	0	1	0	1	1	1
6	0	1	1	0	0	1	0	1
7	0	1	1	1	0	1	0	0
8	1	0	0	0	1	1	0	0
9	1	0	0	1	1	1	0	1
10	1	0	1	0	1	1	1	1
11	1	0	1	1	1	1	1	0
12	1	1	0	0	1	0	1	0
13	1	1	0	1	1	0	1	1
14	1	1	1	0	1	0	0	1
15	1	1	1	1	1	0	0	0

Truth table: Binary input (B) and Gray output (G)



What to notice

Only 1 Gray bit changes from one row to the next. G3 always equals B3.

Conclusion and Applications

1 Outcome

- Converter built with only 3 XOR gates
- All 16 inputs tested in Logisim
- Outputs matched the truth table

2 Applications

- Rotary and shaft encoders
- Karnaugh maps (K-maps)
- Error reduction in digital systems and ADCs

3 Improvements

- Extend to 8 bits or more
- Add a Gray to Binary converter
- Build on hardware (ICs or FPGA)

Take-home: Gray code = Binary XOR the bit above it. Thank you!